



KNN, Isolation Forest, and
all scalars fit on Train only

17 channels: 8 causal values,
8 masks, 1 anomaly flag

Windows built inside each split;
lookback 72, output 144

Horizon-specific valid origins;
H144-common sensitivity

Past Active Power is causal input; future Active Power is target only.

Validation global MSE = total SSE / valid targets; Test is excluded from fitting, training, and checkpoint selection.

Leakage-aware and sample-matched temporal evaluation protocol